

Midterm Examination, Spring-2024

Course Title: Probability & Statistics

Course Code: STAT-2311

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1. a) Define statistics and explain its importance in the field of computer science and engineering.

Ans: **Statistics** is the branch of mathematics that deals with the collection, analysis, interpretation, and presentation of numerical data. It is used to make informed decisions based on data.

Statistics is essential in computer science and engineering, helping in data analysis, algorithm optimization, and decision-making. In **data science and machine learning**, it is used for model training, predictions, and pattern recognition. **Algorithm performance analysis** relies on statistical methods to evaluate efficiency, while **software quality control** uses defect prediction and testing metrics to ensure reliability. In **cybersecurity**, statistics helps detect anomalies and prevent cyberattacks. **Big data analytics** benefits from statistical tools to process and interpret large datasets, while **database optimization** uses statistics to improve query performance. In **computer vision and image processing**, statistical models enable pattern detection and object recognition. Lastly, in **simulation and modeling**, engineers use statistics to predict outcomes and improve system designs. Overall, statistics plays a crucial role in enhancing technology, optimizing processes, and making informed decisions in the field.

b) What is a quantitative variable? Determine whether these variables are discrete, continuous, or qualitative:

- (i) Time it takes for students to finish their assignment
- (ii) Number of books in the library
- (iii) Price of a laptop
- (iv) Number of visitors to a website
- (v) Thesis titles

Ans: A variable that can be measured and expressed numerically is called quantitative variable.

- (i) **Time to finish assignment → Continuous** (Time can take any value, e.g., 2.5 hours).
- (ii) **Number of books in a library → Discrete** (Books can only be counted in whole numbers).
- (iii) **Price of a laptop → Continuous** (Prices can have decimal values, e.g., \$999.99).
- (iv) **Number of visitors to a website → Discrete** (Visitors are counted in whole numbers).
- (v) **Thesis titles → Qualitative** (Thesis titles are categorical, not numerical).

2. a) Define weighted average. Can you compare the weighted average with other types of averages? Write some real-world applications where the weighted average is commonly used.

Ans: **Weighted average** is a type of mean where different values contribute more or less to the final average based on their assigned weights.

The **weighted average** differs from other types of averages because it assigns different levels of importance (weights) to values, whereas other averages treat all values equally.

1. **Weighted Average vs. Simple Average:**

- The simple average (mean) gives equal importance to all values, while the weighted average considers some values as more significant than others.
- Example: A student's GPA is a weighted average because courses with more credit hours have a greater impact on the final GPA, unlike a simple average where all courses would be treated equally.

2. **Weighted Average vs. Median:**

- The median is used when data has extreme values (outliers) since it focuses on the middle value, whereas the weighted average adjusts for significance but can still be affected by outliers.
- Example: When analyzing salaries, the median is more reliable than the weighted average if a few executives earn extremely high salaries.

3. **Weighted Average vs. Mode:**

- The mode is the most frequently occurring value, which is useful for categorical data, while the weighted average applies only to numerical data and considers importance.
- Example: In product sales, the mode helps identify the most popular item, but a weighted average would determine the overall revenue impact of different products.

In summary, the weighted average is particularly useful when some values contribute more than others, making it ideal for cases like academic grading, financial indices, and business analytics. However, other averages are better suited for specific types of data analysis.

Real-World Applications of Weighted Average

1. Academic Grading (GPA Calculation)
2. Stock Market Indices (S&P 500, Dow Jones)
3. Financial Portfolio Management
4. Business and Sales Analysis
5. Sports Performance Metrics
6. Economics (Consumer Price Index - CPI, Inflation Calculation)
7. Manufacturing and Quality Control
8. Medical Research and Drug Testing
9. Employee Performance Evaluation
10. Elections and Polling Analysis

b) In a study, students were surveyed regarding their weekly coding practice time, which was categorized as follows:

Hours Spent (per week)	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	5	8	9	20	12	6

- (i) Calculate the mean and median weekly coding practice time.
- (ii) What percentage of students spend more than 20 hours per week on coding practice?
- (iii) Draw an ogive curve and hence locate the median.

Ans: (i) Mean Weekly Coding Time

Using the formula:

$$\bar{x} = \frac{\sum(f \cdot x)}{\sum f}$$

$$\bar{x} = \frac{(5 \times 2.5) + (8 \times 7.5) + (9 \times 12.5) + (20 \times 17.5) + (12 \times 22.5) + (6 \times 27.5)}{5 + 8 + 9 + 20 + 12 + 6}$$

$$= \frac{12.5 + 60 + 112.5 + 350 + 270 + 165}{60}$$

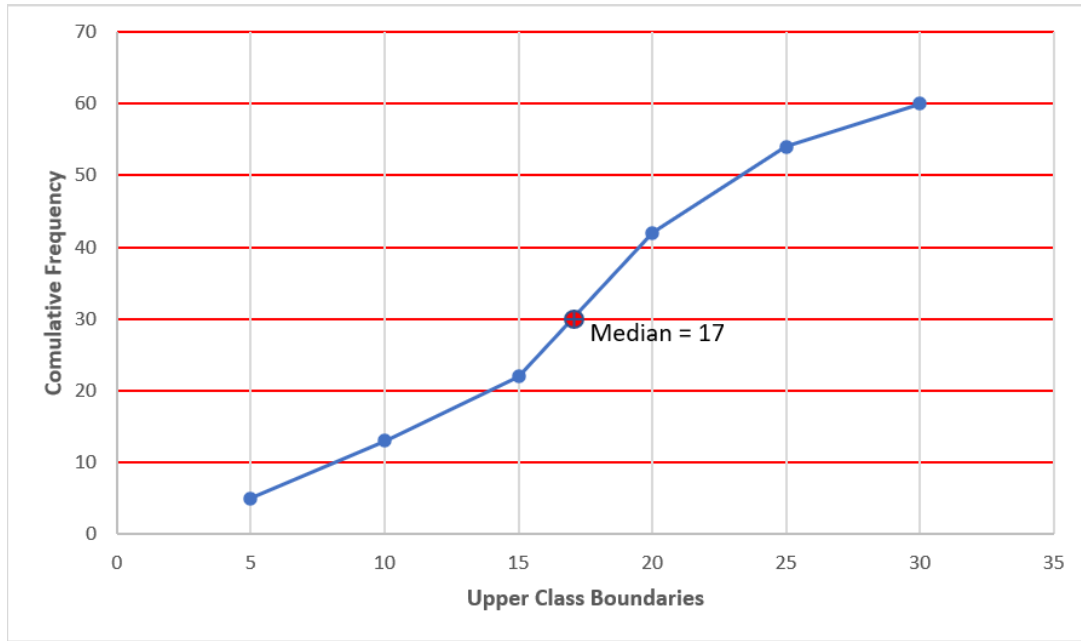
$$= \frac{970}{60} \approx 16.17 \text{ hours}$$

(ii) Percentage of Students Spending More Than 20 Hours

$$\frac{(12 + 6)}{60} \times 100 = \frac{18}{60} \times 100 = 30\%$$

(iii) Cumulative Frequency Table for Ogive Curve:

Hours Spent (per week)	Frequency (f)	Cumulative Frequency (CF)
0 - 5	5	5
5 - 10	8	13
10 - 15	9	22
15 - 20	20	42
20 - 25	12	54
25 - 30	6	60



Or

a) Distinguish between:

- (i) Primary data and secondary data.
- (ii) Discrete variable and continuous variable.

Ans:

(i) Primary Data vs. Secondary Data:

Feature	Primary Data	Secondary Data
Definition	Data collected firsthand by the researcher for a specific purpose.	Data that has already been collected by someone else and is used for another purpose.
Source	Surveys, experiments, interviews, observations, etc.	Government reports, books, articles, online databases, research papers, etc.
Accuracy	Generally more accurate and relevant for the specific study.	May be less accurate as it was collected for a different purpose.
Time & Cost	Time-consuming and expensive to collect.	Less time-consuming and more cost-effective.
Examples	A company conducting a customer satisfaction survey.	Using census data to analyze population growth trends.

(ii) Discrete Variable vs. Continuous Variable:

Feature	Discrete Variable	Continuous Variable
Definition	A variable that takes only specific, countable values.	A variable that can take any value within a given range.
Nature	Countable (finite or infinite but distinct values).	Measurable and can have infinite possible values in a range.
Type of Data	Whole numbers (integers).	Can have decimal or fractional values.
Possible Values	Specific values (e.g., 0, 1, 2, 3...).	Any value within a given range (e.g., 2.5, 3.75, 4.01...).
Examples	Number of students in a class, number of cars in a parking lot, goals scored in a match.	Height, weight, temperature, time taken to complete a task.

b) Suppose the heights (in inches) of students in your section are as follows:

66, 77, 68, 63, 72, 60, 66, 70, 71 and your height in inches. What would be the mean, median, and mode of this data? Can you compute the variance, standard deviation, and coefficient of variation for this dataset? Which measure would best represent this data?

Ans: Given Data:

The heights (in inches) of students in the section are:

66, 77, 68, 63, 72, 60, 66, 70, 71, 65

Measures of Central Tendency:

- **Mean:**

$$\bar{x} = \frac{\sum x_i}{n} = \frac{66 + 77 + 68 + 63 + 72 + 60 + 66 + 70 + 71 + 65}{10} = 67.8$$

- **Median:**

Arranging the data in ascending order:

60, 63, 65, 66, 66, 68, 70, 71, 72, 77

Since there are 10 values, the median is the average of the 5th and 6th values:

$$\text{Median} = \frac{66 + 68}{2} = 67.0$$

- **Mode:**

The most frequently occurring value in the dataset is **66** (appears twice).

Measures of Dispersion:

- **Variance:**

Using the formula for sample variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Substituting the values,

$$s^2 = 23.96$$

- **Standard Deviation:**

$$s = \sqrt{s^2} = \sqrt{23.96} = 4.89$$

- **Coefficient of Variation (CV):**

$$CV = \left(\frac{s}{\bar{x}} \right) \times 100 = \left(\frac{4.89}{67.8} \right) \times 100 = 7.22\%$$

Best Measure to Represent the Data:

Since the data does not have extreme outliers and the standard deviation is moderate, the **mean (67.8 inches)** is the best measure to represent the central tendency of the dataset.

3. a) What is meant by dispersion? What are its important measures? Discuss mean deviation and the coefficient of variation.

Ans:

Dispersion refers to the extent to which data values deviate from the central value (mean, median, or mode). It indicates the degree of variability or spread in a dataset. A high dispersion means that data points are more spread out, while a low dispersion suggests that they are closely clustered around the central value.

Important Measures of Dispersion:

Dispersion can be measured using the following methods:

1. **Range (R):**
 - The difference between the maximum and minimum values in the dataset.
 - Formula: $R = \text{Max Value} - \text{Min Value}$
2. **Mean Deviation (MD):**
 - The average of the absolute deviations of data points from the mean or median.
 - Formula (about the mean): $MD = \frac{\sum |x_i - \bar{x}|}{n}$
3. **Variance (s^2):**
 - The average of the squared deviations from the mean.
 - Formula for sample variance: $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$
4. **Standard Deviation (s):**
 - The square root of the variance, providing a measure of dispersion in the same unit as the data.
 - Formula: $s = \sqrt{s^2}$
5. **Coefficient of Variation (CV):**
 - A relative measure of dispersion that expresses standard deviation as a percentage of the mean.
 - Formula: $CV = \left(\frac{s}{\bar{x}} \right) \times 100$

Mean Deviation:

Mean deviation (also called the average absolute deviation) is a measure of dispersion that calculates the average of the absolute differences between each data point and the mean (or median). It provides a simple way to measure how much the values deviate from the central tendency.

- **Advantages:**
 - Simple to calculate and interpret.
 - Uses absolute values, avoiding cancellation of differences.
- **Disadvantages:**
 - Ignores the direction of deviations.
 - Less commonly used than standard deviation in statistical analysis.

Coefficient of Variation (CV):

The coefficient of variation is a standardized measure of dispersion that allows comparison between datasets with different units or scales.

- **Interpretation:**
 - A **lower CV** indicates less relative variability.
 - A **higher CV** suggests greater relative dispersion in the data.
 - Useful for comparing the relative consistency of two datasets, especially when means differ significantly.

- **Example:**

If dataset A has a mean of 50 and a standard deviation of 5, and dataset B has a mean of 100 with a standard deviation of 10, both will have the same CV:

$$CV_A = \left(\frac{5}{50}\right) \times 100 = 10\%$$

$$CV_B = \left(\frac{10}{100}\right) \times 100 = 10\%$$

This shows that both datasets have the same relative variability.

b) The runs scored by two batsmen in their last 6 ODI cricket matches are given below:

Batsman	Match 1	Match 2	Match 3	Match 4	Match 5	Match 6
X	65	25	78	Sum of last two digits of your ID	51	34
Y	40	35	18	28	25	41

(i) Which batsman has a higher average run?

(ii) Which batsman is more consistent?

Ans: (i) Sum of last two digits of my ID is 5. So, the runs scored by the two batsmen in their last six matches are:

Match	1	2	3	4	5	6
Batsman X	65	25	78	5	51	34
Batsman Y	40	35	18	28	25	41

The mean (average) is calculated using the formula:

$$\bar{x} = \frac{\sum x_i}{n}$$

For Batsman X:

$$\bar{x}_X = \frac{65 + 25 + 78 + 5 + 51 + 34}{6} = \frac{258}{6} = 43.0$$

For Batsman Y:

$$\bar{x}_Y = \frac{40 + 35 + 18 + 28 + 25 + 41}{6} = \frac{187}{6} = 31.17$$

Batsman X has a higher average run.

(ii) To determine consistency, we use the **standard deviation**, which measures the dispersion of runs. A lower standard deviation indicates more consistent performance. The formula for standard deviation is:

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n - 1}}$$

For Batsman X:

$$\begin{aligned} s_X &= \sqrt{\frac{(65 - 43)^2 + (25 - 43)^2 + (78 - 43)^2 + (5 - 43)^2 + (51 - 43)^2 + (34 - 43)^2}{6 - 1}} \\ &= \sqrt{\frac{(22)^2 + (-18)^2 + (35)^2 + (-38)^2 + (8)^2 + (-9)^2}{5}} \\ &= \sqrt{\frac{484 + 324 + 1225 + 1444 + 64 + 81}{5}} \\ &= \sqrt{\frac{3622}{5}} = \sqrt{724.4} = 26.91 \end{aligned}$$

For Batsman Y:

$$\begin{aligned} s_Y &= \sqrt{\frac{(40 - 31.17)^2 + (35 - 31.17)^2 + (18 - 31.17)^2 + (28 - 31.17)^2 + (25 - 31.17)^2 + (41 - 31.17)^2}{6 - 1}} \\ &= \sqrt{\frac{(8.83)^2 + (3.83)^2 + (-13.17)^2 + (-3.17)^2 + (-6.17)^2 + (9.83)^2}{5}} \\ &= \sqrt{\frac{77.99 + 14.68 + 173.46 + 10.06 + 38.06 + 96.64}{5}} \\ &= \sqrt{\frac{410.89}{5}} = \sqrt{82.18} = 9.06 \end{aligned}$$

Since **Batsman Y** has a lower standard deviation (9.06) compared to Batsman X (26.91), **he** is more consistent in his performance.